

Bruce Skarin

25 Herricks Ln, Millbury, MA 01527

bruceskarin@gmail.com • 508-318-8182

Summary

Innovative Data Scientist with over 20 years of experience designing end-to-end simulation models, ML/AI solutions, and advanced analytics to solve complex challenges in insurance, social systems, and operational domains. Expert in developing scalable models for claims processing, text analytics, and customer experience optimization, with a proven track record of reducing administrative burdens through automation. Skilled in leading cross-functional teams, driving inner-source initiatives, and delivering responsible solutions with real-world impact. As a husband and father of two, I'm inspired by family, the outdoors, and a passion for AI, space, and sustainable energy that make me optimistic about technology's power to solve big problems.

Skills

- **Machine Learning & AI:** Classical ML, Generative Models, PyTorch, TensorFlow, Scikit-learn, NLTK, LangChain, Domino Data Lab, SageMaker
- **Modeling & Simulation:** AnyLogic, Vensim, Agent-Based Modeling, System Dynamics
- **Programming & Data:** Python, R, SQL, Java, Snowflake, Hive, Hadoop, Streamlit, Shiny, Quarto, ETL Pipelines, Pandas, NumPy, SciPy, GitLab, CI/CD, Unit Testing
- **Claims Analytics:** Claims Processing, Text Analytics, Process Optimization
- **Leadership:** Cross-Functional Teams, Inner-Source Development, Stakeholder Collaboration
- **Tools:** MS Office, Jira, Slack, VS Code, Git, Copilot, Gemini, ChatGPT, Grok

Experience

Senior Data Scientist, *USAA*

2018-May 2025

- Developed end-to-end ML/AI solutions for claims handling and customer experience, including models to detect at-risk customers and extract insights from unstructured text, improving satisfaction and operational efficiency.
- Led Python inner-source initiatives, creating shared packages for database and object store connections, fostering collaboration across engineering and data science teams.
- Contributed to R packages for data asset connectivity and model risk management documentation using Quarto, driving 10x reductions document creation while ensuring quality and regulatory compliance.
- Developed and delivered Data Scientist onboarding and upskilling initiatives to serve the whole enterprise and mentored junior colleagues.
- Built simulation models from claims data to optimize staffing and planning, reducing administrative burdens through automated forecasting and reporting.
- Designed scalable ETL pipelines using Snowflake, SQL, Hive, and Hadoop, transforming transactional data into analytics-ready datasets for claims operations.
- Delivered production-ready, unit tested, Python code and Shiny/Streamlit interfaces, enabling stakeholders to access real-time intelligence for monthly forecasting and reporting.

Scientist III, *Charles River Analytics, Cambridge, MA*

2015-2017

- Led ML-driven projects for the US Army and Office of Naval Research, developing scalable models for social network analysis and behavior.
- Principal Investigator for CHARISMA: Built ML models (graph networks, text analytics) to detect and classify subversive behavior in social media.
- Principal Investigator for MultiMAST: Designed multi-resolution simulation models (AnyLogic, SpatialOS) for mega-city population dynamics.
- Spearheaded proposals for predictive and agent-based modeling solutions, securing funding for innovative analytics in complex systems.

Bruce Skarin, *continued*

Consultant – Data Sciences Modeling & Simulation, *Private Contractor*, Millbury, MA 2014-2015

- Developed a Java-based agent-based model for DARPA, simulating synthetic social networks derived from Twitter data.
- Optimized model parameters using advanced analytics, delivering scalable solutions for social system simulations.

Senior Research Engineer, *Aptima, Inc.*, Woburn, MA 2004-2014

- Led \$10M+ in funded research (AFRL, ONR, DARPA) for ML and simulation-based solutions, focusing on population dynamics, text analytics, and collaboration optimization.
- Principal Investigator for SCIPR/FACETS: Built country-scale agent-based models (Java, AnyLogic) for opinion change, integrated ML text analytics (dirichlet multinomial regression) to process events from news articles, relevant to novel AI workflows.
- Project Manager for SCALE/CARTOS/SEAMS: Invented a patented method (US9123022B2) for improving collaboration, partnering with MIT Media Lab to analyze communication data, applicable to workflow automation.
- Lead Engineer for MOST: Developed optimization algorithms (Visual Basic, simulated annealing) for wargaming scenarios, enhancing decision-making in resource-constrained environments.

Engineering Department Manager, *Irwin Research and Development*, Yakima, WA 1997-1999

- Managed engineering teams and cross-departmental workflows, leading redesigns for plastics recycling systems with high-volume healthcare applications.
- Utilized CAD for precision designs, optimizing manufacturing processes for scalability and efficiency.

Education

Worcester Polytechnic Institute, Worcester, MA

M.S., System Dynamics, 2011 | B.S., System Dynamics, 2002 (Graduated with Distinction, Provost's MQP Award)

Publications & Patents

- Skarin, B., Duchon, A., Allopena, P., Dejordy, R. (2015). "Systems and methods for analyzing entity profiles" U.S. Patent #: US9123022B2
- Stickgold, E., Skarin, B., Stewart, I., & Lofdahl, C. (2015). Extending Generative Models of Large Scale Networks. Proceedings of the 24th Conference on Behavior Representation in Modeling and Simulation
- Skarin, B., & Melhuish, J. (2015). "US Campaign Finance and Institutional Corruption in Public Policy," Proceedings of the 33rd International Conference of the System Dynamics Society.
- Skarin, B. (2011). "Distributed Group Problem Solving." Proceedings of the 29th International Conference of the System Dynamics Society.
- McCormack, R. Skarin, B., Yohai, I., Pfautz, S., Duchon, A. Freeman, J., & Salter, W. (2011). "Assessing opinion change through the integration of agent-based models and natural language processing." HSCB Focus 2011, 8-10, Chantilly, VA.
- Skarin, B, Weil, S, Dean, C, Duchon, A, Olguin, D, Kim, T, Jaeger, A. (2010) "Teams-of-Teams Performance Assessment through Multi-Channel Interaction Monitoring," Interservice/Industry Training, Simulation, and Education Conference.
- Kott, A. & Skarin, B. (2010) Insurgency and Security. In A. Kott & G. Citrenbaum (Eds.) *Estimating Impact: A handbook of computational methods and models for anticipating economic, social, political, and security effects in international interventions* (pp. 239-262). New York: Springer.
- Skarin, B., Skorinko, J., Saeed, K. & Pavlov O. (2009). "Modeling the Cycles of Gang and Criminal Behavior." Proceedings of the 27th International Conference of the System Dynamics Society.
- Levchuk, G., Skarin, B., Pattipati, K. (2009). "Identifying Critical Resources and Operations of the Adversaries from Incomplete Data". Proceedings of the 14th International Command and Control Research and Technology Symposium.